Financial Engineering

Introducing The Implied Volatility Surface

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Outline of the course

- Implied, Local and Stochastic Volatility
- Fourier pricing - Thomas Kokholm
- Adding Jumps: introduction to Lévy processes and exponential Lévy models
- Combining Jumps with SV via Time Changes
- At this point Thomas will take over for real, and bring you through vol derivatives plus more.

Basic (operative) knowledge of MATLAB is assumed.

Outline

A quick review of the Black-Scholes model

The Black-Scholes model

In the BS model, the price-process of an asset S is described by a GBM

$$
\begin{aligned}
dS_t &= \mu S_t dt + \sigma S_t dZ_t \quad \Rightarrow \\
X_t = \log(S_t) &= X_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma Z_t \quad \Rightarrow \\
X_{t+\Delta t} - X_t &\sim (\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}N(0,1)
\end{aligned}
$$

Implications

- log-returns are independent
- identically distributed
- normally distributed

In particular, the volatility is constant.

Black-Scholes Risk-neutral

Also the **risk-neutral** price process follows a GMB, albeit with $\mu = r - d$, so risk-neutral returns are also normally distributed

$$X_{t+\Delta t} - X_t \sim (r - d - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}N(0,1)$$

Recall that under the Black-Scholes dynamics the price of a call with strike at K and expiry at T is given by the well known BS formula

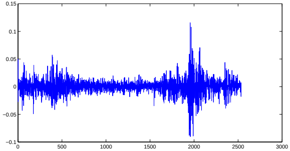
$$e^{-rT}E^Q[(S_T - K)^+] = S_0 e^{-dT}N(d_1) - Ke^{-rT}N(d_2)$$

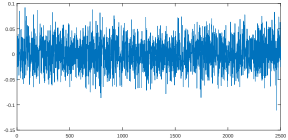
where

$$d_1 = \frac{\log(S_0/K) + (r - d + \sigma^2/2)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}$$

Model vs Empirics -1

This plot shows SPX daily log returns from 02/01/2001 to 02/31/2011 (left) vs i.i.d. returns (right)

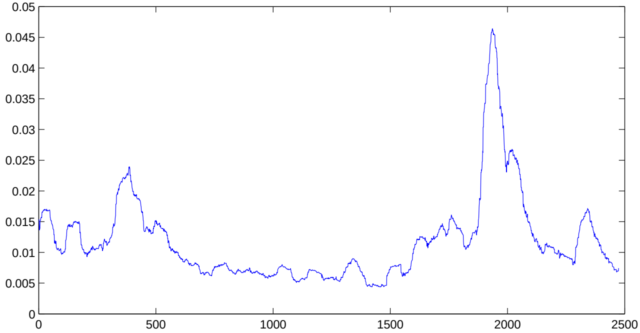




- Volatility doesn't seem to be constant.
- Large moves follow large moves and small moves follow small moves- "volatility clustering"

Model vs Empirics -2

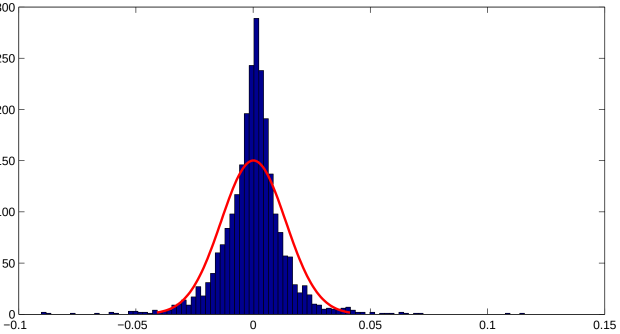
This plot shows SPX daily volatility estimates for the same period



Although naive estimates (60 days-rolling window), clear indication of non-constant vol.

Model vs Empirics - laughing off normality -1

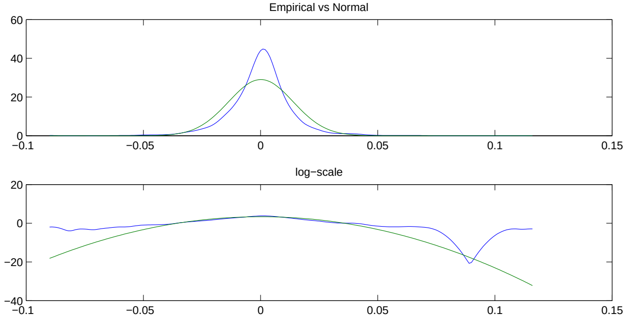
Frequency distribution of (10 years of) SPX daily returns compared with the normal distribution.



Descriptive statistics: mean=0.0136% (annualized=3.42%), STD=1.37% (annualized=21.81%), skewness=0.0846, kurtosis=11.428.

Model vs Empirics - laughing off normality -2

Empirical distribution (kernel density estimator) vs fitted normal.



Stylized features: High central peak (top panel) and fat tails (bottom panel in log-scale).

Mean-Variance Mixtures Distributions: a first look

Mean-Variance Mixtures Distributions: a first look

High central peak and fat tails are characteristics of mean-variance mixtures distributions

$$X \sim \mu + \theta V + \sigma\sqrt{V} \cdot N(0,1), \quad V \perp N(0,1) \tag{1}$$

where

- μ (location), θ (skewness) and σ are real parameters with $\sigma > 0$;
- V is a distribution on the positive real line (representing the random variance);
- V is independent of the standard normal $N(0,1)$.

Gamma Distribution

A popular choice for V is the gamma distribution $\Gamma(\mu, \nu)$ with mean value μ and variance ν, i.e. with density $f_V(x)$ given by

$$f_V(x) = \frac{\left(\frac{\mu}{\nu}\right)^{\frac{\mu^2}{\nu}}}{\Gamma(\frac{\mu^2}{\nu})} \; x^{\frac{\mu^2}{\nu}-1} \exp\left(-\frac{\mu}{\nu}x\right) \qquad x > 0 \tag{2}$$

where $\Gamma(x)$ is the gamma function.

The characteristic function (c.f.) $\phi_V(u) = E[\exp(iuV)]$ is given by

$$\phi_V(u) = \left(1 - iu\frac{\nu}{\mu}\right)^{-\frac{\mu^2}{\nu}} \tag{3}$$

Remark: Watch out for different parameterizations in the literature!

Variance Gamma Distribution

By choosing $V \sim \Gamma(1, \nu)$ (for parameter identification), the resulting mixed distribution $X \sim \mu + \theta V + \sigma\sqrt{V} \cdot N(0, 1)$ is known as the Variance-Gamma distribution with parameters μ, σ, ν, θ. It has density $f_X(x)$

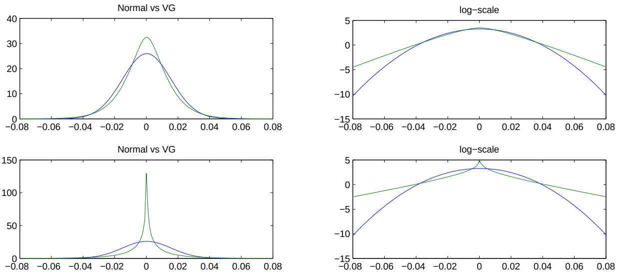
$$
\begin{aligned}
f_X(x) = & \frac{2\exp\left(\theta(x-\mu)/\sigma^2\right)}{\sigma\sqrt{2\pi}\nu^{1/\nu}\Gamma(1/\nu)} \cdot \left(\frac{|x-\mu|}{\sqrt{2\sigma^2/\nu+\theta^2}}\right)^{1/\nu-1/2} \times \\
& K_{1/\nu-1/2}\left(\frac{|x-\mu|\sqrt{2\sigma^2/\nu+\theta^2}}{\sigma^2}\right) \qquad -\infty < x + \infty
\end{aligned}
\tag{4}
$$

where $K_\eta(\cdot)$ is a modified Bessel function of the third kind (again, ambiguity in terminology: MATLAB calls it modified Bessel function of the SECOND kind).
The (much simpler) c.f. $\phi_X(u) = E[\exp(iuX)]$ is given by

$$
\phi_X(u) = e^{i\mu u}\left(1 - i\theta\nu u + (\sigma^2\nu u^2)/2\right)^{-\frac{1}{\nu}} \tag{5}
$$

VG vs Normal

Fixing $\theta = 0$, comparison between VG and Normal distribution, with the same mean=0.0136% and STD=1.37%



Top panel $\nu = 0.5$, bottom panel $\nu = 3$.

So what have we found out?

- The normal distribution is not a good fit to observed returns due to fat tails.
- Mean-variance mixture distributions, such as VG, obtained by randomizing variance in a normal display such a property.
- How about fitting the VG to the SPX returns? You may consider taking a look at it.
- Constant volatility is an absurd assumption. But are mean-variance mixture distributions the full answer to it? Think about it.

- Now all this is information coming from **historical/time-series preliminary analysis.** That is this is information/critic of BS model under the physical/objective7statistical measure.

- How about empirical evidence against BS assumptions coming from **liquid derivative prices?**

- Focussing on the returns, is the normality assumption valid in the risk-neutral world?

Calibration

Calibration: fitting the model to derivatives market-prices

- Assume you are dealing with a parametric derivative pricing model, such as the Black-Scholes model or a more general model.
- How do we fit/estimate the parameter values?
- Rather than relying on estimates based on historical time series of the underlying, the problem of estimating the parameter values is typically tackled via a procedure known as **Calibration**.
- Calibration involves finding values of parameters such that the model is able to reproduce (as close as possible) the prices of the "calibration instruments", i.e., liquid derivative products observed in the market.
- Once the model is calibrated, it can be used to price illiquid instruments (barriers, asians etc. etc.)

The calibration procedure may be schematically described as follows:

- Select from the market the calibration instruments. Typically, this is a panel of calls (and/or puts) $C_{mkt}(K_i, T_j)$ with strikes K_i,$i = 1, \ldots, m$, and maturities T_j, $j = 1, \ldots, n$.

- Let θ denote the parameter set of the underlying model and let $C_{mod}(K_i, T_j; \theta)$ denote the model prices for the calibration instruments.

- The calibration consists in finding the parameter values θ^* which match the observed prices as close as possible according to a given error function, i.e.,

$$\theta^* = \arg \inf_{\theta} error(C_{mod}(K_i, T_j; \theta) - C_{mkt}(K_i, T_j), i, j)$$

- Typical error function are the Least Square Errors, where one minimizes

$$error = \sum_{i,j} \left(C_{mod}(K_i, T_j; \theta) - C_{mkt}(K_i, T_j) \right)^2$$

- or the Mean Absolute Percentage Error

$$error = \sum_{i,j} \frac{|C_{mod}(K_i, T_j; \theta) - C_{mkt}(K_i, T_j)|}{C_{mkt}(K_i, T_j)}$$

Calibration of the Black-Scholes model

- In the BS model, the only unobservable parameter is the volatility σ.

- In this example, we fix a given day and we collect a set of market prices for calls $C_{mkt}(K_i, T_j)$ corresponding to different strikes K_i and maturities T_j: for the same K_i, T_j we also have BS formula for the call price

$$C_{BS}(K, T, \sigma) = e^{-dT} S_0 N(d_1) - Ke^{-rT} N(d_2)$$

with $d_{1/2} = \frac{\log(S_0 e^{(r-d)T}/K)}{\sigma\sqrt{T}} \pm \frac{\sigma\sqrt{T}}{2}$.

- The calibrated parameter σ^* is the parameter value which minimizes the sum of square errors:

$$\sigma^* = \arg\min_{\sigma} \sum_{i,j} \left(C_{BS}(K_i, T_j, \sigma) - C_{mkt}(K_i, T_j)\right)^2$$

Calibration results

- In spreadsheet "SPXData" you may find SPX data for July, 31st, 2009
- The resulting value is $\sigma^* = 0.2624$ and the corresponding fit of BS to prices is as follows:

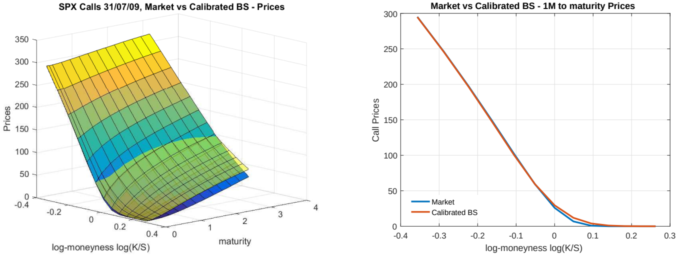


Figure 1: BS calibrated prices vs market prices: whole surface (left panel) and 1 month slice (right panel)

- It doesn't look all that bad - especially at the short maturity of 1 month.
- However, just looking at dollar prices might not provide the full picture of the fit.
- Just as percentage errors can be more insightful of absolute errors - or the log-scale can be useful to analyze the densities tail-behaviour, it might be a good idea to change the scale of the call prices surface to something else.
- That something else is known as the **implied volatility surface**.

Implied Volatility

Moving to the Implied Volatility "scale"

We now introduce an alternative way to quote options in substitution to raw dollar prices and we start with the fundamental definition.

Definition (Implied volatility of a call option)

Let $C(S_0, K, T)$ be today's price (free of arbitrage) of a call with strike K and maturity T written on a stock paying dividends at dividend yield d and today's price S_0.

Then the **Implied Volatility** $\sigma_{BS}(S_0, K, T)$ of such a call is defined as the volatility value to be plugged in the Black-Scholes formula to match the call price $C(S_0, K, T)$, i.e., $\sigma_{BS}(S_0, K, T)$ is implicitly defined as the solution of the following equation

$$C(S_0, K, T) = C_{BS}(S_0, K, \sigma_{BS}(S_0, K, T), T)\,. \tag{6}$$

where

$$C_{BS}(S_0, K, \sigma, T) = S_0 e^{-dT} N(d_1) - K e^{-rT} N(d_2)$$

In other words:

- With a slight abuse of notation, fix S_0, K, and T and let $C_{BS}(\sigma)$ denote the Black-Scholes call formula as function of the volatility parameter only.

- The implied volatility of the given call $C(S_0, K, T)$ is defined as

$$\sigma_{BS}(S_0, K, T) = C_{BS}^{-1}(C(S_0, K, T))$$

where C_{BS}^{-1} is the inverse function of $C_{BS}(\sigma)$.

- By the very definition, the implied volatility of a BS call with parameter σ is σ for every given strike K or maturity T

$$\sigma = C_{BS}^{-1}(C_{BS}(S_0, K, \sigma, T)).$$

- If a given call price $C(S_0, K, T)$ is within the no arbitrage bounds, then its implied volatility is a well defined quantity.

The implied volatility is well defined

Exercise

Fix S_0, K, and T and let $C_{BS}(\sigma)$ be defined as above. Show the following

1. $C_{BS}(\cdot) : (0, +\infty) \to \left(\max(S_0 e^{-dT} - Ke^{-rT}, 0) \, , \, S_0 \right)$.

2. $C_{BS}(\cdot)$ is an invertible function. (Compute the so-called Vega $\partial C_{BS} / \partial \sigma$).

3. Conclude that if a call price $C(S_0, K, T) \in \max(S_0 e^{-dT} - Ke^{-rT}, 0)$, then its implied volatility is well defined.

The implied volatility surface

- Given an arbitrage free surface of option prices $C(S_0, K, T)$ - either from a model or from the market - the **volatility surface** $\sigma_{BS}(S_0, K, T)$ is the associated surface of implied volatilities.

- Implied volatilities are in one-to-one correspondence with option prices.

- They can be seen as a useful alternative way to quote options instead of raw prices:option prices can be compared across different strikes, maturities and underlyings.

Calibration results, in the implied volatility scale

- For the BS model, the implied volatilities are constant and equal to σ for every K and T;

- Transforming the prices presented in figure (1) to implied volatilities, the calibrated BS model ($\sigma^* = 0.2624$) gives the following fit to market implied volatilities

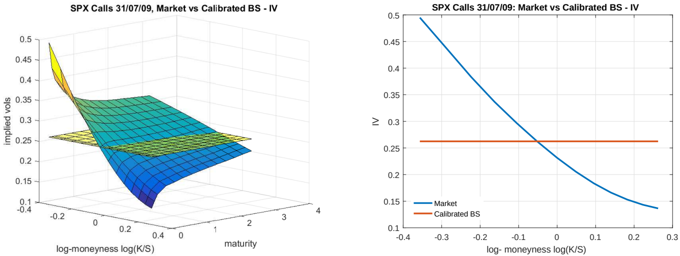


Figure 2: BS calibrated prices vs market prices: whole surface (left panel) and 1 month slice (right panel)

If BS was a good model for market prices, we should observe a flat vol surface!

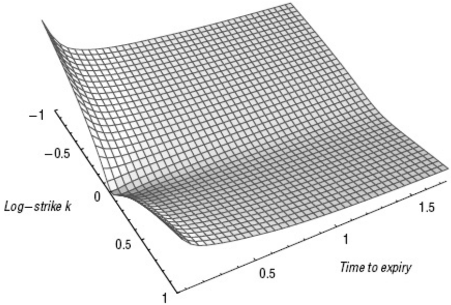


FIGURE 3.2 Graph of the SPX-implied volatility surface as of the close on September 15, 2005, the day before triple witching.

Figure 3: Implied Vol Surface observed on 15/09/2005

- Typical behavior across moneyness $\log(K/S)$ known as smile (hockey-stick) effect. Notice negative skew at-the-money.

- Rubinstein (1994) suggests that the over-pricing of out of the money puts is an indication of crash-o-phobia:

Volatility smiles and crash-o-phobia

- Literally, "crash-o-phobia" means fear of a market crash.
- The U.S. stock market crash of October 1987 was in fact the event marking the appearance of smile patters in volatilities implied by equity options.
- Put options are used as hedging instruments to protect against large downward movements in stock prices.
- This demand by investors due to portfolio insurance strategies increases the price of put options and therefore the left tail of the implied distribution has more weight
- A number of studies (as well as our dataset) suggest that the crash-o-phobia is not only a short term worry, but also a long term concern.

How to generate a smile? An experiment.

Let's fix the maturity T. By risk-neutral evaluation, we have that the price of a call is given by

$$C(S_0, K, T) = e^{-rT} E^Q[(e^{X_T} - K)^+] \tag{7}$$

In the BS model

$$X_T = \log S_1 = \log S_0 + r - q - 1/2\sigma^2 T + \sqrt{\sigma^2 T} N(0,1).$$

- Inspired by the return analysis, we could consider σ^2 as a random variable , $\sigma^2 \perp N(0,1)$, and model X_T as a mean-variance mixture as introduced in eq. (1). For example, we could choose $(\sigma^2) \sim \Gamma(m, \nu)$ to obtain the so-called VG model introduced by Carr& Madan & Chang (1998) (albeit re-parametrization).
- Prices in the new model are still given by the expectation in (7), for which we have a semi-explicit formula (recall VG density (4)).

Evaluating calls in the VG model by MC

Evaluation by MC of (7) is straightforward:

1. Generate a large number n of random numbers $V^1, \dots, V^n$ from $\Gamma(m, \nu)$;

2. Generate n standard normal random numbers $\varepsilon^1, \dots, \varepsilon^n$ from $N(0, 1)$;

3. Compute the corresponding log-prices

$$X_T^i = \log S_0 + r - q - 1/2 V^i T + \sqrt{V^i T} \varepsilon^i$$

   The call price C_{VG} in the VG model is then given by the arithmetic average

$$C_{VG}(S_0, K, T) = e^{-rT} \frac{1}{n} \sum_{i=1}^{n} (e^{X_T^i} - K)^+$$

The associated implied vol function

And now the experiment goes as follows:

- Fix a maturity (e.g. $T = 1$, choose two parameters m and ν for the gamma r.v. and, using the same random numbers, compute $C_{VG}(S_0, K_j, T)$ for a set of strikes K_j, $j = 1, \ldots, m$.

- Numerically invert (MATLAB function available)

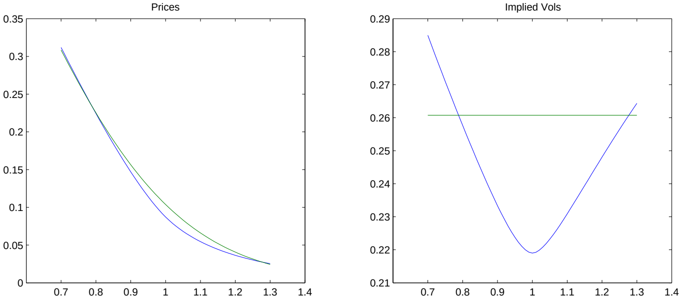
$$C_{VG}(S_0, K_j, T) = C_{BS}(S_0, K_j, \sigma_{BS}(S_0, K_j, T), T)$$

  to find the implied volatilities.

- plot them against K.

A smile!

And the result is as follows (green line BS prices corresponding to $\sigma = 26\%$)



We do manage to obtain a smile!!! We can conclude that randomizing the variance of a normal distribution injects curvature in an otherwise flat implied vol function.